

Pediatric Anemia Diagnosis Aide - Current-State Product Specification

As-built scope, clinical evidence, functional behavior, data model, algorithm, architecture, verification, and production gaps

Authoritative as-built baseline for product, clinical, engineering, quality, regulatory, research, and diligence review

PREPARED FOR	Nick Miethe
DATE	2026-07-16
STATUS	Specification v1.0 for app v0.2.0 plus the Algorithm Explorer enhancement; research prototype
SCOPE	Current implemented behavior and production gaps; future commercialization strategy is documented separately

Important: This specification documents the current research prototype. It does not establish clinical validation, regulatory status, production security, or suitability for live patient care.

Document control

Field	Value
Specification version	1.0
Specification date	2026-07-16
Application package version	0.2.0
Knowledge-base version	0.1.0-2026-07-15
Evidence reviewed through	2026-07-15
Algorithm Explorer update	Applied to current working build; adds explanatory content and live walkthrough, not new clinical rules
Status	Research prototype; not clinically validated; not cleared or approved as a medical device
Intended audience for this specification	Product, clinical, informatics, engineering, quality, regulatory, research, and diligence teams

SCOPE OF THIS FILE

This is an as-built specification. Future commercial recommendations are in the separate commercialization strategy report. Items labeled production gaps are not current capabilities.

1. Product overview

The Pediatric Anemia Diagnosis Aide is a deterministic, evidence-linked clinical decision-support research prototype for licensed clinicians evaluating pediatric anemia. It collects demographic, CBC, history, examination, smear, and targeted laboratory inputs; derives normalized clinical facts; executes inspectable JSON rules; merges and ranks diagnostic patterns using fixed ordinal priority; and returns safety alerts, rationale, adaptive questions, suggested confirmatory workup, citations, and a reproducible rule audit.

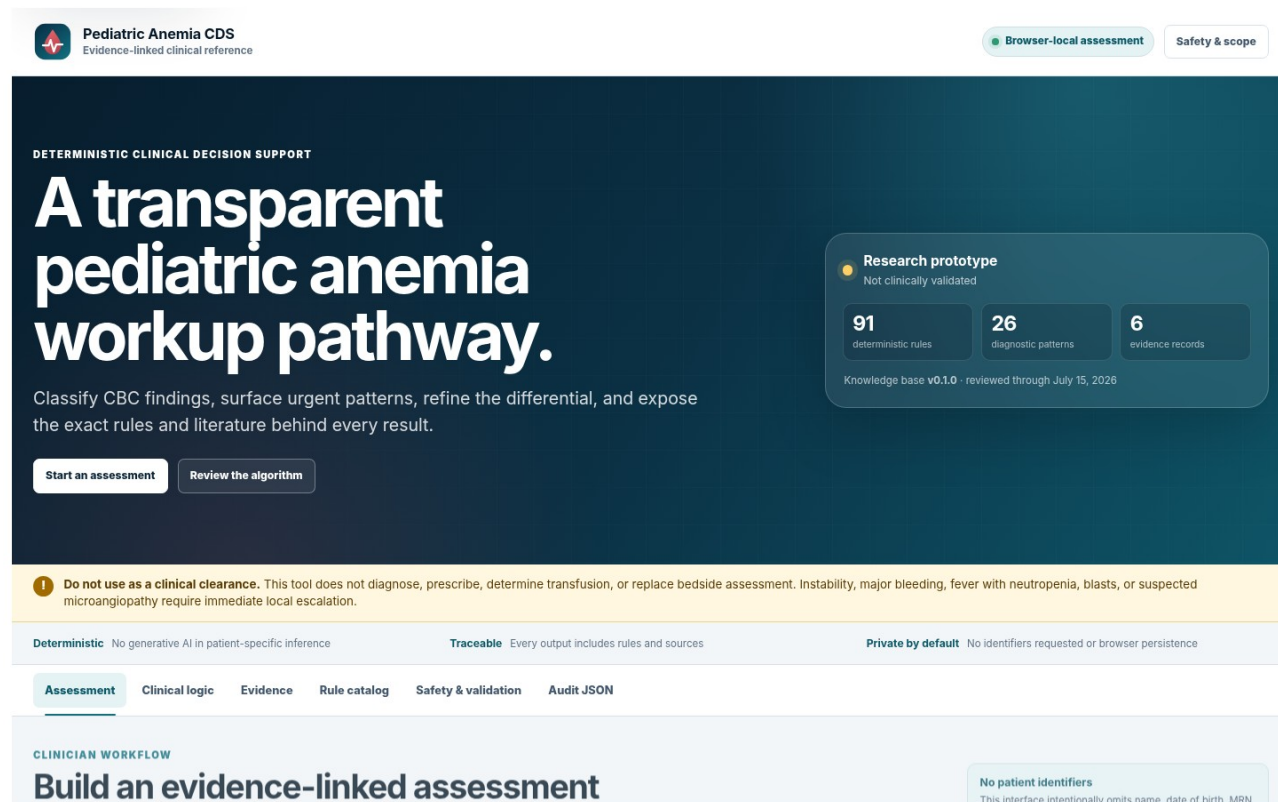


Figure 1. Current clinician workspace: public, browser-local assessment with explicit research-prototype and no-identifiers warnings.

1.1 Product goals

- Help clinicians classify pediatric anemia using local or age/sex fallback reference intervals.
- Surface high-risk findings before etiologic refinement.
- Guide a morphology- and reticulocyte-based differential through explicit deterministic logic.
- Keep missing data visible and ask high-yield follow-up questions.
- Expose the exact rules, evidence IDs, and reasoning behind each result so a clinician can independently review it.
- Provide a reproducible, versioned assessment artifact without generative AI in patient-specific inference.

1.2 Explicit non-goals

- Autonomous diagnosis or replacement of bedside assessment.
- Treatment selection, medication dosing, iron prescription, transfusion instruction, or order placement.
- A validated probability, likelihood ratio, calibrated confidence, sensitivity, or specificity estimate.
- General neonatal anemia classification below 6 months using static fallback thresholds.
- Adult anemia interpretation without supplied adult/local laboratory ranges.
- Persistent patient record, EHR integration, production PHI handling, or enterprise audit service.

2. Intended use, users, and clinical scope

Dimension	As-built specification
Intended user	Licensed healthcare professional; the interface is not designed as independent patient or caregiver CDS.
Intended purpose	Review CBC and related findings and provide transparent diagnostic patterns, alerts, questions, and confirmatory options for clinician consideration.

Dimension	As-built specification
Patient population	Built-in fallback ranges for age 6 months to <18 years.
Young infant behavior	Age <6 months is routed outside the static general-pediatric classification and requires neonatal/young-infant specialist and local reference data.
Adult behavior	Age ≥18 requires adult/local laboratory limits; the pediatric fallback table is not applied.
Care setting	Research/demo use in outpatient, urgent, emergency, education, or referral contexts; no validated deployment setting.
Inference type	Deterministic, rule-based, inspectable, reproducible.
Output type	Ranked diagnostic patterns and follow-up options, not a guaranteed diagnosis.

CLINICAL BOUNDARY

The current software may display emergency or urgent flags, but it is not validated for autonomous time-critical triage, and the UI instructs users to use local escalation for instability, bleeding, fever with neutropenia, blasts, and suspected microangiopathy.

3. Clinical evidence baseline

The evidence review prioritized publications from 2021 through July 2026, with older material retained only when foundational and not replaced in the focused review. The current evidence registry contains six records.

Evidence ID	Source	Role in current logic
AAP2026_IDA	AAP Pediatrics clinical report, 2026 [C1]	Age/sex CBC fallback intervals; ferritin thresholds; inflammation caveats; iron, thalassemia, and inflammatory differentiation.
WHO2024_HB	WHO hemoglobin cutoff guideline, 2024 [C2]	Age/population-aware cutoff governance and transparent provenance.
BLOOD2022_PED_ANEMIA	Blood review, 2022 [C3]	Morphology and reticulocyte framework; broad pediatric differential; smear, hemolysis, marrow, nutrient, renal, endocrine, infectious, inherited causes.
CDC2025_LEAD	CDC lead guidance, updated 2025 [C4]	3.5 micrograms/dL reference value, capillary confirmation, escalation tiers, and iron deficiency consideration.
FDA2026_CDS	FDA CDS guidance, 2026 [R1]	Transparency, intended-use, specificity, time-criticality, and clinician independent-review design constraints.
BSH2020_G6PD	BSH G6PD laboratory guideline, 2020 [C5]	Quantitative testing and false-normal timing/transfusion cautions; labeled foundational exception.

3.1 Encoded fallback CBC intervals

Local laboratory ranges take precedence. The following AAP-derived fallbacks are used only within the encoded age scope and when the required demographic input is available. [C1]

Age	Female Hb lower	Male Hb lower	Female MCV (fL)	Male MCV (fL)	Female RDW upper	Male RDW upper
6 to <24 mo	11.0	11.0	73.3-83.2	71.1-82.2	15.4	15.9
2 to <6 y	11.0	11.0	75.2-85.0	74.1-84.3	14.5	14.7
6 to <12 y	11.2	11.3	78.3-87.7	77.8-86.5	13.9	13.7
12 to <18 y	11.4	12.4	80.5-91.8	80.4-90.1	14.6	13.7

Hemoglobin is in g/dL, MCV in fL, and RDW in percent. Ferritin is interpreted against ≤ 20 ng/mL for children 6 months to <12 years and ≤ 30 ng/mL for adolescents and all menstruating patients, with inflammation caveats. [C1]

4. Functional capabilities

ID	Capability	As-built behavior
FR-001	Scope and warning display	Prominently identifies research-prototype status, licensed-clinician intent, no clinical clearance, and local escalation needs.
FR-002	Core patient and CBC capture	Captures age, sex assigned at birth, menstruation, transfusion, altitude, Hb, MCV, RDW, RBC, WBC, ANC, platelets, flags, and local ranges.
FR-003	Immediate safety screen	Captures instability, major bleeding, respiratory/cardiac/neurologic symptoms, fever, jaundice, dark urine, fatigue, and pallor.
FR-004	Reticulocyte and iron/inflammation capture	Captures response or absolute reticulocytes, ferritin, CRP, transferrin saturation, TIBC/transferrin, serum iron, and sTfR/ferritin index.
FR-005	Specialized laboratory capture	Captures hemolysis, DAT, lead/specimen, hemoglobin analysis, globin tests, G6PD timing, nutrients, renal, thyroid, and liver interpretation.
FR-006	History/exposure questionnaire	Captures dietary, bleeding, inflammatory, renal, infectious, medication, toxin, travel, inherited, congenital, and prior-response factors.
FR-007	Smear and exam capture	Captures target cells, spherocytes, schistocytes, bite/blister cells, sickle cells, stippling, hypersegmentation, blasts, teardrops, nucleated RBCs, elliptocytes, organomegaly, lymphadenopathy, petechiae, and bruising.
FR-008	Reference-range selection	Uses supplied local limits first; otherwise selects encoded AAP age/sex fallback when in scope.
FR-009	Fact derivation	Normalizes input into anemia presence, morphology, RDW, reticulocyte, iron, inflammation, hemolysis, cytopenia, lead, and other rule facts.
FR-010	Rule execution	Evaluates every machine-readable condition and records matched/not-matched state.
FR-011	Pattern merge and rank	Merges rule support into 26 diagnostic patterns and sorts by deterministic level and points; ranking is explicitly not probability.
FR-012	Safety output	Returns emergency, urgent, and important alerts before the differential.
FR-013	Adaptive follow-up	Returns the highest-priority unanswered questions according to the active branch.
FR-014	Evidence and rationale	Shows supporting/conflicting findings, rule IDs, evidence chips, and suggested confirmatory steps.
FR-015	Audit export	Exposes normalized input, result, KB version, and evaluated rule trace as JSON; supports copy/download.

ID	Capability	As-built behavior
FR-016	Worked examples	Loads six de-identified example cases spanning IDA, thalassemia, inflammation, hemolysis, lead, and marrow red flags.
FR-017	Algorithm Explorer	Expands six algorithm stages with data points, expressions, thresholds, studies, examples, branches, and live walkthrough.
FR-018	Evidence and rule catalog	Displays evidence registry and searchable deterministic rules.
FR-019	Print/copy summary	Provides print and copy controls for the clinician-facing result.
FR-020	Prototype REST API	Exposes health, knowledge-base summary, and deterministic assessment endpoints.

Classify CBC findings, surface urgent patterns, refine the differential, and expose the exact rules and literature behind every result.

[Start an assessment](#) [Review the algorithm](#)

Do not use as a clinical clearance. This tool does not diagnose, prescribe, determine transfusion, or replace bedside assessment. Instability, major bleeding, fever with neutropenia, blasts, or suspected microangiopathy require immediate local escalation.

Deterministic No generative AI in patient-specific inference **Traceable** Every output includes rules and sources **Private by default** No identifiers requested or browser persistence

[Assessment](#) [Clinical logic](#) [Evidence](#) [Rule catalog](#) [Safety & validation](#) [Audit JSON](#)

CLINICIAN WORKFLOW

Build an evidence-linked assessment

Use de-identified clinical data only. Local laboratory intervals take precedence over built-in fallback ranges.

Load a worked example

Start blank [Load](#) Core data 3/3 19 supplemental findings entered [Clear form](#)

Clinician review required
Core data complete. Supplemental findings improve discrimination. [Run assessment](#)

Local laboratory intervals override the built-in AAP fallback ranges.

Age (months) 20	Sex assigned at birth Female	Hemoglobin (g/dL) 8.4	MCV (fL) 64
RDW (%) 12	RBC (×10¹²/L) 4.1	WBC (×10⁹/L) 9	ANC (×10⁹/L) 9

Deterministic assessment
Research prototype—not clinically validated - 0.1.0-2026-07-15

ANEMIA
Present

HEMOGLOBIN
8.4 g/dL
Lower limit 11 g/dL

MCV
64 fL
Range 73.3–83.2 fL

RETICULOCYTES
Inappropriately Normal

RANGE SOURCE
AAP2026_IDA
6 to <24 months

Worked example loaded and assessed.

[Copy cited summary](#) [Print](#)

Figure 2. Current worked assessment output: classification, safety, ranked pattern differential, rationale, citations, and next questions.

5. Input data model

The JSON input contract contains eight top-level sections. Most fields are optional so the engine can represent incomplete assessments, although the current symptom/history/exam maps use booleans and therefore do not yet distinguish false from unknown unless the UI/context does so explicitly.

Section	Representative fields
patient	ageMonths; sexAtBirth; menstruating; recentTransfusion; highAltitude
cbc	hemoglobin; mcv; rdw; rbc; wbc; anc; platelets; rbcInterpretation; localRanges; localFlags
reticulocytes	response; absolute

Section	Representative fields
symptoms	instability; respiratory distress; syncope; altered mental status; chest pain; heart failure signs; bleeding; fever; neurologic/renal symptoms; jaundice; dark urine; fatigue; pallor
history	diet/milk; prematurity; malabsorption; menstrual/GI/nasal/other loss; inflammatory/renal/liver/thyroid/infectious disease; viral illness; travel; family/known hemoglobinopathy or hemolysis; medication/toxin/lead; congenital anomalies; prior iron response
exam	splenomegaly; hepatomegaly; lymphadenopathy; petechiae; unexplained bruising
labs	ferritin/CRP/TSAT/TIBC/iron/index; bilirubin/LDH/haptoglobin/DAT; lead and specimen; HbA2/Bart/sickling/globin tests; G6PD; B12/folate/copper; creatinine; TSH; liver tests
smear	target cells; spherocytes; schistocytes; bite/blister cells; sickle cells; basophilic stippling; hypersegmented neutrophils; blasts; teardrops; nucleated RBCs; elliptocytes

5.1 Data quality behavior

- Local ranges override fallback ranges.
- Assay-dependent tests often use local interpretation states (low/normal/high/unknown) rather than hard-coded universal numbers.
- Recent transfusion, acute hemolysis, inflammation, and assay timing can generate interpretive cautions.
- Outside-scope ages are not forced through pediatric fallback thresholds.
- Current production gap: replace boolean maps with explicit present/absent/unknown/not-assessed and include source, timestamp, method, and provenance.

6. Deterministic algorithm specification

Current deterministic assessment pipeline

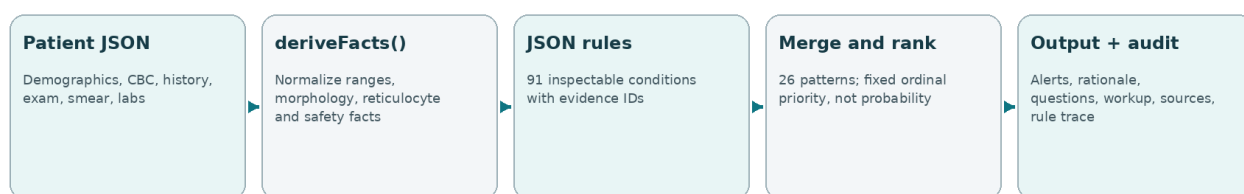


Figure 3. Current software pipeline. The same input and knowledge-base version produce the same result.

6.1 Stage sequence

Stage	Logic	Representative output
1. Establish scope and urgency	Validate age/units/ranges; evaluate instability, bleeding, cytopenias, blasts, schistocytes, lead urgency, and other safety rules	Out-of-scope note or tiered alert.
2. Confirm anemia	Compare Hb to local lower limit or applicable AAP fallback	Present / absent / indeterminate and effective lower limit/source.

Stage	Logic	Representative output
3. Classify morphology	Compare MCV and RDW to local or fallback ranges	Microcytic / normocytic / macrocytic / indeterminate; RDW status.
4. Classify marrow response	Use clinician-entered reticulocyte response or available absolute count/context	High / low / inappropriately normal / unknown pathway.
5. Apply morphology-specific rules	Use iron/inflammation, globin, lead, loss/hemolysis, production, nutrient/endocrine/liver, smear, history, and exam facts	Matched candidate rules and adaptive questions.
6. Merge, rank, and expose provenance	Merge candidate evidence; keep highest evidence label; sort by fixed points; attach evidence, knowns/unknowns, and workup	Ranked pattern list and complete rule audit.

6.2 Iron-deficiency rule core

```

ferritin threshold = 30 ng/mL if menstruating or age 12 to <18 years
                    else 20 ng/mL for age 6 months to <12 years

anemia + ferritin at/below threshold -> iron deficiency anemia pattern
no anemia + ferritin at/below threshold -> iron deficiency without anemia pattern
CRP elevated + ferritin not low -> iron deficiency remains not excluded
sTfR/log10(ferritin) > 2 -> supports iron deficiency
sTfR/log10(ferritin) < 1 + inflammation -> supports anemia of inflammation

```

The app does not automatically prescribe iron, infer adherence, or use a normal/high ferritin during inflammation to exclude iron deficiency. [C1]

6.3 High-reticulocyte hemolysis core

- Ask about overt or occult blood loss.
- A general biochemical hemolysis pattern requires at least two of elevated indirect bilirubin, elevated LDH, or low haptoglobin.
- DAT positive supports immune hemolysis; spherocytes with DAT negative support hereditary spherocytosis; bite/blister cells plus oxidant exposure support G6PD pathway; schistocytes trigger microangiopathic/mechanical concern; sickling evidence supports sickling hemoglobinopathy; fever plus malaria exposure supports an infectious pathway. [C3; C5]

6.4 Low-reticulocyte and macrocytic core

- Renal dysfunction, inflammation, viral context, age, other cell lines, blasts, organomegaly, and congenital findings route among production-limited and marrow pathways.
- Macrocytosis with high reticulocytes redirects toward loss/hemolysis; low/inappropriate reticulocytes plus B12/folate/copper, hypersegmentation, TSH, liver tests, medication, or congenital features refine the branch. [C3]

6.5 Ranking semantics

Label	Meaning
Meets defined pattern	An explicit biochemical, molecular, or source-defined criterion was met.
Strongly supported	A relatively specific combination of findings was met.
Supported	Multiple compatible findings are present, but meaningful alternatives remain.

Label	Meaning
Possible	A lower-specificity pattern warrants a question or test.
Not excluded	A confounder or missing value prevents exclusion.

IMPORTANT

The internal point total only sorts candidates within deterministic levels. It has no probabilistic interpretation.

7. Knowledge-base inventory

Artifact	Count / status
Deterministic rules	91 total: 55 differential, 17 adaptive-question, 13 safety, 6 interpretive-note
Safety severities	5 emergency, 4 urgent, 4 important
Diagnostic patterns	26
Evidence records	6
Worked clinical examples	6
Automated engine tests	10

7.1 Diagnostic-pattern catalog

#	Pattern
1	Iron deficiency anemia
2	Iron deficiency without anemia
3	Anemia of inflammation / inflammatory iron restriction
4	Thalassemia trait or other globin-production pattern
5	Beta-thalassemia pattern
6	Alpha-thalassemia pattern
7	Lead exposure-associated anemia
8	Sideroblastic or iron-loading microcytosis
9	Blood loss
10	Hemolysis, etiology not yet specified
11	Autoimmune hemolytic anemia
12	Hereditary spherocytosis
13	G6PD deficiency
14	Microangiopathic hemolysis
15	Sickling hemoglobinopathy
16	Malaria or parasitic hemolysis
17	Renal anemia

#	Pattern
18	Transient erythroblastopenia of childhood
19	Parvovirus aplastic crisis
20	Marrow failure or infiltration
21	Vitamin B12 or folate deficiency
22	Copper deficiency
23	Thyroid-, liver-, or medication-associated macrocytosis
24	Inherited marrow failure
25	Mixed anemia or transfusion effect
26	Iron-refractory iron deficiency anemia / rare iron handling disorder

8. User interface and workflow

Tab / area	Current behavior
Assessment	Six-stage intake, worked examples, live de-identified snapshot, input-depth indicator, sticky run action, independent results pane.
Clinical logic / Algorithm Explorer	Six expanded stages with explanations, data fields, expressions, thresholds, evidence links, examples, morphology branches, and live engine walkthrough.
Evidence	Displays evidence registry, source metadata, review recency, and links.
Rule catalog	Searchable 91-rule catalog with IDs, categories, output behavior, priorities, and evidence references.
Safety and validation	Research status, limitations, clinical boundaries, and production gate.
Audit JSON	Normalized input, deterministic output, KB version, matched/evaluated rules, copy and download.
Responsive behavior	Desktop sidebar/two-column workflow and compact mobile navigation; print-friendly result.

8.1 Algorithm Explorer six stages

- Establish scope and urgency.
- Confirm anemia.
- Classify red-cell morphology.
- Classify marrow response.
- Apply morphology-specific rules.
- Rank and expose provenance.

The Explorer update changes educational and explanatory presentation only. It does not alter the 91-rule clinical knowledge base.

9. Technical architecture

Layer / file	Responsibility
index.html; styles.css; site-overrides.css	Responsive clinician SPA, tabs, forms, cards, print layout, warnings.

Layer / file	Responsibility
src/app.js	UI state, field mapping, examples, assessment rendering, copy/print/audit actions.
src/algorithmExplorer.js	Expanded algorithm explanations and live walkthrough.
src/facts.js	Input normalization and derived clinical facts.
src/ruleEngine.js	Generic JSON condition evaluator, candidate merge, deterministic sorting.
src/engine.js	Assessment contract and evidence-linked result assembly.
src/referenceRanges.js	AAP fallback interval selection and range provenance.
src/evidence.js	Evidence registry loading and citation resolution.
data/rules.json	91 deterministic rules.
data/candidates.json	26 merged pattern definitions and default next steps.
data/evidence.json	Versioned source metadata.
data/reference-ranges.json	Machine-readable fallback ranges.
data/algorithm-explainers.json	Educational stage explanations, examples, equations, and source IDs.
schemas/*.json	Patient input, assessment output, rule, and candidate contracts.
server.mjs; openapi.yaml	Static server and prototype REST API.
scripts/; tests/	Static build, knowledge-base validation, smoke test, and deterministic engine tests.

9.1 Runtime behavior

- The browser downloads static JavaScript and JSON assets and runs the assessment locally.
- The browser interface does not call the included API and does not automatically persist entered data.
- The Node API mirrors the deterministic engine for integration testing and returns results without intentional body logging or persistence.
- Opening index.html directly from disk fails because browsers block local fetch of JSON; use the included server or static host.
- Node.js 20 or later is specified for local server/build/test scripts.

9.2 REST API

Method	Path	Purpose
GET	/health	Health and knowledge-base version.
GET	/api/v1/knowledge-base	Evidence registry and rule/pattern counts.
POST	/api/v1/assess	Validate and execute a deterministic assessment.

10. Deployment and security posture

Mode	As-built support	Limitations
Static hosting	Publish dist/ to Cloudflare Pages, GitHub Pages, Netlify, Vercel, S3/CloudFront, or equivalent	No API; security headers depend on host; public site must not be treated as approved PHI environment.

Mode	As-built support	Limitations
Node server	Static site plus prototype REST endpoints	No authentication, tenant isolation, durable audit, production observability, BAA, or PHI governance.
Docker	Containerized Node server	Containerization alone does not establish security, privacy, compliance, or clinical readiness.
Browser-local	No automatic server transmission or persistence from assessment UI	Browser/device, extensions, screenshots, printing, and hosting still require a threat model.

10.1 Existing privacy-oriented design

- No name, date of birth, medical record number, or contact fields in the interface.
- No third-party fonts, scripts, analytics, or trackers in the packaged site.
- robots.txt requests that search engines not index the prototype.
- Assessment values remain browser-local in the current UI and are not automatically saved.

10.2 Production security gaps

- Authentication, SSO, RBAC, MFA, tenant isolation, authorization, key management, encryption-at-rest, audited access, retention/deletion, backup/restore, incident response, vulnerability management, and penetration testing are not implemented.
- No HIPAA or FTC applicability assessment has been completed for a production business model. [R7-R8]
- No production secure-SDLC, SBOM governance, quality system, or regulated change-control process is implemented.

11. Verification status

Check	Current result
Engine tests	10 of 10 passed.
Knowledge-base validation	91 rules, 26 candidates, and 6 evidence records validated.
Static build	Passed; current Algorithm Explorer assets included.
Live server/API smoke test	Passed; expected health and deterministic assessment response.
Browser testing	Desktop and mobile interaction checks previously passed for example loading, assessment rendering, citations, navigation, print, and audit.
Interpretation	Verifies software behavior and packaging only; does not establish clinical validity, safety, diagnostic accuracy, or utility.

11.1 Current test scenarios

- AAP fallback interval by age and sex.
- Local ranges override built-in ranges.
- Iron deficiency anemia example reaches the defined pattern.
- Beta-thalassemia example ranks a beta-thalassemia pattern.
- Inflammatory example ranks anemia of inflammation.
- DAT-negative spherocytic hemolysis includes hereditary spherocytosis.

- Marrow red flags trigger emergency and urgent alerts.
- Elevated capillary lead triggers confirmation and lead pathway.
- Young infant is not forced through built-in thresholds.
- Identical input produces identical output.

12. Known limitations and production gates

Limitation	Required production gate
No clinical validation	Dual specialist review, multi-site retrospective validation, prospective silent mode, and controlled rollout.
Incomplete neonatal logic	Dedicated neonatal research, postnatal/gestational model, local ranges, and specialist validation before inclusion.
Assay and lab dependence	UCUM units, method metadata, laboratory medicine review, site-specific mapping and validation.
Boolean missingness	Explicit tri-state or four-state answers, source, timestamp, and completeness checks.
Rare disease resolution	Specialist confirmation and explicit limits for marrow, molecular, membrane, enzyme, and hemoglobinopathy diagnoses.
No probability model	Retain ordinal language unless a separately validated probabilistic model is developed and calibrated.
Regulatory status unknown	Final intended-use/claims analysis, specialized counsel, and possible FDA interaction. [R1]
No production content operations	Exact source passages, evidence grade, dual approval, expiry, surveillance, semantic diffs, signed releases, rollback.
No production security/privacy	Enterprise security architecture, contracts, threat model, controls, testing, and legal assessment. [R7-R8]
Content reuse rights not resolved	Copyright and license review for guideline tables, excerpts, and derived content.

RELEASE PROHIBITION

Do not use the current prototype for live patient care or market it as clinically validated until the production gates are completed and documented.

13. Recommended acceptance criteria for the next controlled release

- Every rule has exact source passage, page/section, applicability, effective date, review date, and two clinical approvals.
- Every rule has at least one positive and one negative activation test; every numeric cutoff has boundary tests; dangerous pathways have explicit miss tests.
- All clinical booleans use explicit present/absent/unknown/not-assessed semantics.
- The knowledge package is immutable, signed, reproducible, semantically diffed, and rollback-capable.
- An external pediatric hematologist, general pediatrician, and laboratory medicine reviewer approve the module.
- Retrospective and silent-mode protocol, reference standard, statistical plan, subgroup plan, and incident process are approved.

- Intended use, labeling, privacy/security architecture, and regulatory assessment are approved for the target deployment.

14. Source package inventory

The verified working package contains 73 files and is approximately 742 KB before screenshots or report artifacts. Key deliverables include the web interface, source modules, JSON knowledge base, schemas, examples, tests, documentation, Node server, static build, Dockerfile, OpenAPI contract, deployment headers, and release verification.

```
pediatric-anemia-site/
  index.html, styles.css, site-overrides.css
  src/{app,algorithmExplorer,engine,facts,ruleEngine,referenceRanges,evidence}.js
  data/{rules,candidates,evidence,reference-ranges,algorithm-explainers}.json
  schemas/{patient-input,assessment-output,rule,candidate}.schema.json
  examples/*.json
  tests/engine.test.mjs
  scripts/{build-static,validate-kb,smoke-test}.mjs
  server.mjs, openapi.yaml, Dockerfile, _headers, robots.txt
  docs/*.md
  dist/
```

References

- [C1] American Academy of Pediatrics. Prevention, Screening, Diagnosis, and Treatment of Iron Deficiency and Iron Deficiency Anemia in Infants, Children, and Adolescents: Clinical Report. 2026. <https://publications.aap.org/pediatrics/article/158/1/e2026077414/207901/Prevention-Screening-Diagnosis-and-Treatment-of> *Primary current clinical source in the app.*
- [C2] World Health Organization. Guideline on haemoglobin cutoffs to define anaemia in individuals and populations. 2024. <https://www.who.int/publications/i/item/9789240088542> *Supports age- and population-aware cutoff governance.*
- [C3] Blood / American Society of Hematology. Anemia in the pediatric patient. 2022. <https://doi.org/10.1182/blood.2021013018> *Recent peer-reviewed morphology and reticulocyte framework.*
- [C4] US Centers for Disease Control and Prevention. Recommended Actions Based on Blood Lead Level. 2025. <https://www.cdc.gov/lead-prevention/hcp/clinical-guidance/index.html> *Current lead reference value, confirmation, and escalation pathways.*
- [C5] British Society for Haematology. Laboratory diagnosis of G6PD deficiency: A British Society for Haematology Guideline. 2020. <https://pubmed.ncbi.nlm.nih.gov/?term=Laboratory+diagnosis+of+G6PD+deficiency+British+Society+for+Haematology> *Foundational exception outside the five-year priority window.*
- [R1] US Food and Drug Administration. Clinical Decision Support Software: Guidance for Industry and FDA Staff. 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software> *Final guidance, January 2026.*
- [R2] World Health Organization. SMART Guidelines. Current page accessed 2026. <https://www.who.int/teams/digital-health-and-innovation/smart-guidelines> *Describes the challenge of translating guidance into digital systems.*
- [R3] HL7 International. SMART App Launch Implementation Guide v2.2.0. Current published version accessed 2026. <https://hl7.org/fhir/smart-app-launch/> *FHIR-based app launch and authorization standard.*
- [R4] HL7 International. CDS Hooks Implementation Guide v2.0.1. Current published version accessed 2026. <https://cds-hooks.hl7.org/> *Workflow-triggered CDS integration standard.*
- [R5] HL7 International. FHIR PlanDefinition R4. R4. <https://hl7.org/fhir/R4/plandefinition.html> *Shareable conditions, triggers, and executable actions.*

- [R6]** HL7 International. FHIR Questionnaire R5. R5. <https://www.hl7.org/fhir/questionnaire.html> *Structured, ordered clinical data capture and QuestionnaireResponse.*
- [R7]** US Department of Health and Human Services. The HIPAA Security Rule. Page reviewed March 2026.
<https://www.hhs.gov/hipaa/for-professionals/security/index.html> *Administrative, physical, and technical safeguards for ePHI.*
- [R8]** US Federal Trade Commission. Health Breach Notification Rule. Final rule notice 2024.
<https://www.ftc.gov/legal-library/browse/rules/health-breach-notification-rule> *May apply to vendors of personal health records and related entities.*